

87. Living Arrangements In 2012, there were 73,817 (in thousands) children under the age of 18 living in the United States. Of these children, 50,267 lived with both parents, 2924 lived with their father only, and 2634 lived with neither parent. How many children lived with their mother only? *Source: U.S. Census.*

88. Sales Calls Suppose that Kendra Gallegos has appointments with 9 potential customers. Kendra will be ecstatic if all 9 of these potential customers decide to make a purchase from her. Of course, in sales there are no guarantees. How many different sets of customers may place an order with Kendra? (*Hint: Each set of customers is a subset of the original set of 9 customers.*)

89. Electoral College U.S. presidential elections are decided by the Electoral College, in which each of the 50 states, plus the District of Columbia, gives all of its votes to a candidate.* Ignoring the number of votes each state has in the Electoral College, but including all possible combinations of states that could be won by either candidate, how many outcomes are possible in the Electoral College if there are two candidates? (*Hint: The states that can be won by a candidate form a subset of all the states.*)

90. Musicians A concert featured a cellist, a flutist, a harpist, and a vocalist. Throughout the concert, different subsets of the four musicians performed together, with at least two musicians playing each piece. How many subsets of at least two are possible?

91. Games In David Gale's game of Subset Takeaway, the object is for each player, at his or her turn, to pick a nonempty proper subset of a given set subject to the condition that no subset chosen earlier by either player can be a subset of the newly chosen set. The winner is the last person who can make a legal move. Consider the set $A = \{1, 2, 3\}$. Suppose Joe and Dorothy are playing the game and Dorothy goes first. If she chooses the proper subset $\{1\}$, then Joe cannot choose any subset that includes the element 1. Joe can, however, choose $\{2\}$ or $\{3\}$ or $\{2, 3\}$. Develop a strategy for Joe so that he can always win the game if Dorothy goes first. *Source: Scientific American.*

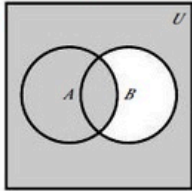
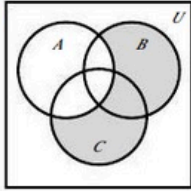
States In the following list of states, let $A = \{\text{states whose name contains the letter } e\}$, let $B = \{\text{states with a population}$

of more than 4,000,000}, and $C = \{\text{states with an area greater than 40,000 square miles}\}$. *Source: The World Almanac and Book of Facts 2013.*

State	Population (1000s)	Area (sq. mi.)
Alabama	4803	52,419
Alaska	723	663,267
Colorado	5117	104,094
Florida	19,058	65,755
Hawaii	1375	10,931
Indiana	6517	36,418
Kentucky	4369	40,409
Maine	1328	35,385
Nebraska	1843	77,354
New Jersey	8821	8721

- 92. a.** Describe in words the set $A \cup (B \cap C)'$.
b. List all elements in the set $A \cup (B \cap C)'$.
- 93. a.** Describe in words the set $(A \cup B)' \cap C$.
b. List all elements in the set $(A \cup B)' \cap C$.

YOUR TURN ANSWERS

1. {Ohio, Oklahoma, Oregon} 2. True 3. $2^4 = 16$ subsets
 4. {2, 4, 8} 5. {1, 3, 5, 6, 7, 9, 11, 12}
 6.  7. 

8. 23 9. 18

*The exceptions are Maine and Nebraska, which allocate their electoral college votes according to the winner in each congressional district.

12.2

Introduction to Probability

APPLY IT

What is the probability that a death in the United States was caused by heart disease or cancer?

After introducing probability, we will answer this question in Example 13.

When a physician prescribes an antibiotic, she specifies the *exact* amount of the drug that you should receive. On the other hand, the managing pharmacist is faced with the problem of ordering the medicine. The pharmacist may have a good estimate of the amount of an antibiotic that will be sold during the day, but it is impossible to predict the *exact* amount. The quantity of the antibiotic that customers will purchase during a day

is *random*: it cannot be predicted exactly. There are many problems that come up in life science applications of mathematics that involve random phenomena—those for which *exact* prediction is impossible. The best that we can do is to determine the *probability* of the possible outcomes.

Sample Spaces In probability, an **experiment** is an activity or occurrence with an observable result. Each repetition of an experiment is called a **trial**. The possible results of each trial are called **outcomes**. The set of all possible outcomes for an experiment is the **sample space** for that experiment. A sample space for the experiment of tossing a coin is made up of the outcomes heads (h) and tails (t). If S represents this sample space, then

$$S = \{h, t\}.$$

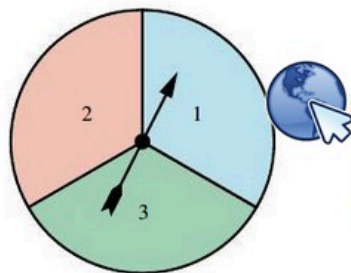


FIGURE 14

EXAMPLE 1 Sample Spaces

Give the sample space for each experiment.

- (a) A spinner like the one in Figure 14 is spun.

SOLUTION The three outcomes are 1, 2, or 3, so the sample space is

$$S = \{1, 2, 3\}.$$

- (b) For the purposes of a public opinion poll, respondents are classified as young, middle-aged, or senior, and as male or female.

SOLUTION A sample space for this poll could be written as a set of ordered pairs:

$$S = \{(\text{young, male}), (\text{young, female}), (\text{middle-aged, male}), (\text{middle-aged, female}), (\text{senior, male}), (\text{senior, female})\}.$$

- (c) An experiment consists of studying the numbers of boys and girls in families with exactly 3 children. Let b represent *boy* and g represent *girl*.

SOLUTION A three-child family can have 3 boys, written bbb , 3 girls, ggg , or various combinations, such as bgg . A sample space with four outcomes (not equally likely) is

$$S_1 = \{3 \text{ boys, } 2 \text{ boys and } 1 \text{ girl, } 1 \text{ boy and } 2 \text{ girls, } 3 \text{ girls}\}.$$

Notice that a family with 3 boys or 3 girls can occur in just one way, but a family of 2 boys and 1 girl or 1 boy and 2 girls can occur in more than one way. If the *order* of the births is considered, so that bgb is different from gbg or ggb , for example, another sample space is

$$S_2 = \{bbb, bbg, bgb, gbb, bgg, gbg, ggb, ggg\}.$$

The second sample space, S_2 , has equally likely outcomes if we assume that boys and girls are equally likely. This assumption, while not quite true, is approximately true, so we will use it throughout this book. The outcomes in S_1 are not equally likely, since there is more than one way to get a family with 2 boys and 1 girl (bbg , gbg , or gbb) or a family with 2 girls and 1 boy (ggb , gbg , or bgg), but only one way to get 3 boys (bbb) or 3 girls (ggg).

TRY YOUR TURN 1

YOUR TURN 1 Two coins are tossed, and a head or a tail is recorded for each coin. Give a sample space where each outcome is equally likely.

CAUTION

An experiment may have more than one sample space, as shown in Example 1(c). The most convenient sample spaces have equally likely outcomes, but it is not always possible to choose such a sample space.

Events An **event** is a subset of a sample space. If the sample space for tossing a coin is $S = \{h, t\}$, then one event is $E = \{h\}$, which represents the outcome “heads.”

An ordinary die is a cube whose six different faces show the following numbers of dots: 1, 2, 3, 4, 5, and 6. If the die is fair (not “loaded” to favor certain faces over others), then any one of the faces is equally likely to occur when the die is rolled. The sample space for the experiment of rolling a single fair die is $S = \{1, 2, 3, 4, 5, 6\}$. Some possible events are listed below.

The die shows an even number: $E_1 = \{2, 4, 6\}$.

The die shows a 1: $E_2 = \{1\}$.

The die shows a number less than 5: $E_3 = \{1, 2, 3, 4\}$.

The die shows a multiple of 3: $E_4 = \{3, 6\}$.

Using the notation introduced earlier in this chapter, notice that $n(S) = 6$, $n(E_1) = 3$, $n(E_2) = 1$, $n(E_3) = 4$, and $n(E_4) = 2$.



EXAMPLE 2 Events

For the sample space S_2 in Example 1(c), write the following events.

- (a) Event H : The family has exactly two girls.

SOLUTION Families with three children can have exactly two girls with either bgg , gbg , or ggb , so event H is

$$H = \{bgg, gbg, ggb\}.$$

- (b) Event K : The three children are the same sex.

SOLUTION Two outcomes satisfy this condition: all boys or all girls.

$$K = \{bbb, ggg\}$$

- (c) Event J : The family has three girls.

SOLUTION Only ggg satisfies this condition, so

$$J = \{ggg\}.$$

TRY YOUR TURN 2

In Example 2(c), event J had only one possible outcome, ggg . Such an event, with only one possible outcome, is a **simple event**. If event E equals the sample space S , then E is called a **certain event**. If event $E = \emptyset$, then E is called an **impossible event**.



EXAMPLE 3 Events

Suppose a coin is flipped until both a head and a tail appear, or until the coin has been flipped four times, whichever comes first. Write each of the following events in set notation.

- (a) The coin is flipped exactly three times.

SOLUTION This means that the first two flips of the coin did not include both a head and a tail, so they must both be heads or both be tails. Because the third flip is the last one, it must show the side of the coin not yet seen. Thus the event is

$$\{hht, tth\}.$$

- (b) The coin is flipped at least three times.

SOLUTION In addition to the outcomes listed in part (a), there is also the possibility that the coin is flipped four times, which only happens when the first three flips are all heads or all tails. Thus the event is

$$\{hht, tth, hhhh, hhht, tttt, ttth\}.$$

YOUR TURN 2 Two coins are tossed, and a head or a tail is recorded for each coin. Write the event E : The coins show exactly one head.

- (c) The coin is flipped at least two times.

SOLUTION This event consists of the entire sample space:

$$S = \{ht, th, hht, tth, hhhh, hhht, tttt, ttth\}.$$

This is an example of a certain event.

- (d) The coin is flipped fewer than two times.

SOLUTION The coin cannot be flipped fewer than two times under the rules described, so the event is the empty set $\emptyset$. This is an example of an impossible event.

Since events are sets, we can use set operations to find unions, intersections, and complements of events. A summary of the set operations for events is given below.

Set Operations for Events

Let E and F be events for a sample space S .

$E \cap F$ occurs when both E **and** F occur;

$E \cup F$ occurs when E **or** F **or both** occur;

E' occurs when E does **not** occur.



EXAMPLE 4 Biology Majors

A study of biology majors grouped such students into various categories, which can be interpreted as events when a student is selected at random. Consider the following events:

E : student is under 20;

F : student is white;

G : student is female.

Describe the following events in words.

- (a) E'

SOLUTION E' is the event that the student is 20 or over.

- (b) $F \cap G'$

SOLUTION $F \cap G'$ is the event that the student is white and not a female; that is, the student is a white male.

- (c) $E \cup G$

SOLUTION $E \cup G$ is the event that the student is under 20 or is female. Note that this event includes all students under 20, both male and female, and all female students of any age.

TRY YOUR TURN 3

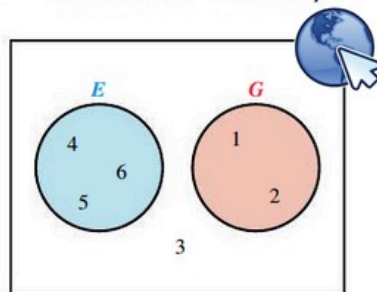
YOUR TURN 3 In Example 4, describe the following event in words: $E' \cap F'$.

Two events that cannot both occur at the same time, such as rolling an even number and an odd number with a single roll of a die, are called *mutually exclusive events*.

Mutually Exclusive Events

Events E and F are **mutually exclusive events** if $E \cap F = \emptyset$.

Any event E and its complement E' are mutually exclusive. By definition, mutually exclusive events are disjoint sets.



$$E \cap G = \emptyset$$

FIGURE 15

EXAMPLE 5 Mutually Exclusive Events

Let $S = \{1, 2, 3, 4, 5, 6\}$, the sample space for tossing a single die. Let $E = \{4, 5, 6\}$, and let $G = \{1, 2\}$. Then E and G are mutually exclusive events since they have no outcomes in common: $E \cap G = \emptyset$. See Figure 15.

Probability For sample spaces with *equally likely* outcomes, the probability of an event is defined as follows.

Basic Probability Principle

Let S be a sample space of equally likely outcomes, and let event E be a subset of S . Then the **probability** that event E occurs is

$$P(E) = \frac{n(E)}{n(S)}.$$

By this definition, the probability of an event is a number that indicates the relative likelihood of the event.

CAUTION The basic probability principle applies only when the outcomes are equally likely.

**EXAMPLE 6** Basic Probabilities

Suppose a single fair die is rolled. Use the sample space $S = \{1, 2, 3, 4, 5, 6\}$ and give the probability of each event.

(a) E : The die shows an even number.

SOLUTION Here, $E = \{2, 4, 6\}$, a set with three elements. Since S contains six elements,

$$P(E) = \frac{3}{6} = \frac{1}{2}.$$

(b) F : The die shows a number less than 10.

SOLUTION Event F is a certain event, with

$$F = \{1, 2, 3, 4, 5, 6\},$$

so

$$P(F) = \frac{6}{6} = 1.$$

(c) G : The die shows an 8.

SOLUTION This event is impossible, so

$$P(G) = 0.$$

TRY YOUR TURN 4

YOUR TURN 4 In Example 6, find the probability of event H : The die shows a number less than 5.

A standard deck of 52 cards has four suits: hearts ($\heartsuit$), clubs ($\clubsuit$), diamonds ($\diamondsuit$), and spades ($\spadesuit$), with 13 cards in each suit. The hearts and diamonds are red, and the spades and clubs are black. Each suit has an ace (A), a king (K), a queen (Q), a jack (J), and cards numbered from 2 to 10. The jack, queen, and king are called *face cards* and for many purposes can be thought of as having values 11, 12, and 13, respectively. The ace can be thought of as the low card (value 1) or the high card (value 14). See Figure 16. We will refer to this standard deck of cards often in our discussion of probability.

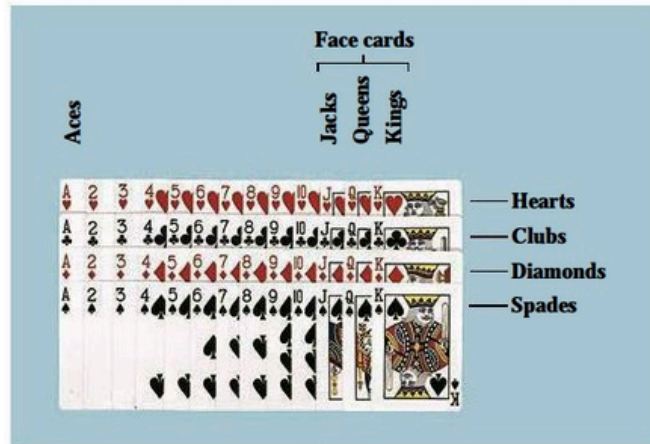


FIGURE 16

**EXAMPLE 7** Playing Cards

If a single playing card is drawn at random from a standard 52-card deck, find the probability of each event.

(a) Drawing an ace

SOLUTION There are 4 aces in the deck. The event “drawing an ace” is
 $\{\text{heart ace, diamond ace, club ace, spade ace}\}.$

Therefore,

$$P(\text{ace}) = \frac{4}{52} = \frac{1}{13}.$$

(b) Drawing a face card

SOLUTION Since there are 12 face cards (three in each of the four suits),

$$P(\text{face card}) = \frac{12}{52} = \frac{3}{13}.$$

(c) Drawing a spade

SOLUTION The deck contains 13 spades, so

$$P(\text{spade}) = \frac{13}{52} = \frac{1}{4}.$$

(d) Drawing a spade or a heart

SOLUTION Besides the 13 spades, the deck contains 13 hearts, so

$$P(\text{spade or heart}) = \frac{26}{52} = \frac{1}{2}.$$

TRY YOUR TURN 5

YOUR TURN 5 Find the probability of drawing a jack or a king.

In the preceding examples, the probability of each event was a number between 0 and 1. The same thing is true in general. Any event E is a subset of the sample space S , so $0 \leq n(E) \leq n(S)$. Since $P(E) = n(E)/n(S)$, it follows that $0 \leq P(E) \leq 1$. Note that a certain event has probability 1 and an impossible event has probability 0, as seen in Example 6.

For any event E , $0 \leq P(E) \leq 1$.

The Union Rule To determine the probability of the union of two events E and F in a sample space S , use the union rule for sets,

$$n(E \cup F) = n(E) + n(F) - n(E \cap F),$$

which was proved in Section 12.1. Assuming that the events in the sample space S are equally likely, divide both sides by $n(S)$, so that

$$\frac{n(E \cup F)}{n(S)} = \frac{n(E)}{n(S)} + \frac{n(F)}{n(S)} - \frac{n(E \cap F)}{n(S)}$$

$$P(E \cup F) = P(E) + P(F) - P(E \cap F).$$

Although our derivation is valid only for sample spaces with equally likely events, the result is valid for any events E and F from any sample space, and is called the **union rule for probability**.

Union Rule for Probability

For any events E and F from a sample space S ,

$$P(E \cup F) = P(E) + P(F) - P(E \cap F).$$



EXAMPLE 8 Probabilities with Dice

Suppose two fair dice are rolled. Find each probability.

- (a) The first die shows a 2, or the sum of the results is 6 or 7.

SOLUTION The sample space for the throw of two dice is shown in Figure 17, where 1-1 represents the event “the first die shows a 1 and the second die shows a 1,” 1-2 represents “the first die shows a 1 and the second die shows a 2,” and so on. Let A represent the event “the first die shows a 2,” and B represent the event “the sum of the results is 6 or 7.” These events are indicated in Figure 17. From the diagram, event A has 6 elements, B has 11 elements, the intersection of A and B has 2 elements, and the sample space has 36 elements. Thus,

$$P(A) = \frac{6}{36}, \quad P(B) = \frac{11}{36}, \quad \text{and} \quad P(A \cap B) = \frac{2}{36}.$$

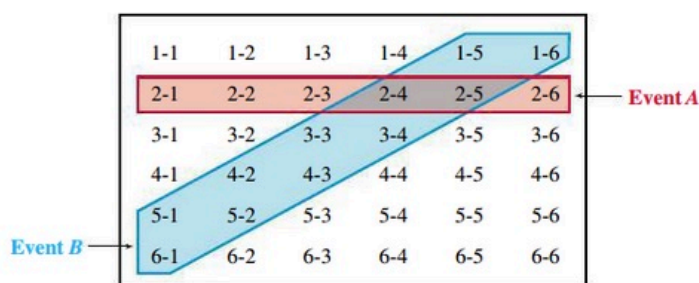


FIGURE 17

By the union rule,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = \frac{6}{36} + \frac{11}{36} - \frac{2}{36} = \frac{15}{36} = \frac{5}{12}.$$

- (b) The sum of the results is 11, or the second die shows a 5.

SOLUTION $P(\text{sum is 11}) = 2/36$, $P(\text{second die shows a 5}) = 6/36$, and $P(\text{sum is 11 and second die shows a 5}) = 1/36$, so

$$P(\text{sum is 11 or second die shows a 5}) = \frac{2}{36} + \frac{6}{36} - \frac{1}{36} = \frac{7}{36}.$$

YOUR TURN 6 In Example 8, find the probability that the sum is 8, or both die show the same number.

TRY YOUR TURN 6

CAUTION

You may wonder why we did not use $S = \{2, 3, 4, 5, \dots, 12\}$ as the sample space in Example 8. Remember, we prefer to use a sample space with equally likely outcomes. The outcomes in set S above are not equally likely—a sum of 2 can occur in just one way, a sum of 3 in two ways, a sum of 4 in three ways, and so on, as shown in Figure 17.

If events E and F are mutually exclusive, then $E \cap F = \emptyset$ by definition; hence, $P(E \cap F) = 0$. In this case the union rule simplifies to $P(E \cup F) = P(E) + P(F)$.

CAUTION

The rule $P(E \cup F) = P(E) + P(F)$ is only valid when E and F are mutually exclusive. When E and F are *not* mutually exclusive, use the rule $P(E \cup F) = P(E) + P(F) - P(E \cap F)$.

The Complement Rule By the definition of E' , for any event E from a sample space S ,

$$E \cup E' = S \quad \text{and} \quad E \cap E' = \emptyset.$$

Since $E \cap E' = \emptyset$, events E and E' are mutually exclusive, so that

$$P(E \cup E') = P(E) + P(E').$$

However, $E \cup E' = S$, the sample space, and $P(S) = 1$. Thus

$$P(E \cup E') = P(E) + P(E') = 1.$$

Rearranging these terms gives the following useful rule for complements.

Complement Rule

$$P(E) = 1 - P(E') \quad \text{and} \quad P(E') = 1 - P(E)$$

**EXAMPLE 9** Complement Rule

If two fair dice are rolled, find the probability that the sum of the numbers rolled is greater than 3. Refer to Figure 17.

SOLUTION To calculate this probability directly, we must find the probabilities that the sum is 4, 5, 6, 7, 8, 9, 10, 11, or 12 and then add them. It is much simpler to first find the probability of the complement, the event that the sum is less than or equal to 3.

$$\begin{aligned} P(\text{sum} \leq 3) &= P(\text{sum is 2}) + P(\text{sum is 3}) \\ &= \frac{1}{36} + \frac{2}{36} \\ &= \frac{3}{36} = \frac{1}{12} \end{aligned}$$

Now use the fact that $P(E) = 1 - P(E')$ to get

$$\begin{aligned} P(\text{sum} > 3) &= 1 - P(\text{sum} \leq 3) \\ &= 1 - \frac{1}{12} = \frac{11}{12}. \end{aligned}$$

TRY YOUR TURN 7

YOUR TURN 7 Find the probability that when two fair dice are rolled, the sum is less than 11.

Odds Sometimes probability statements are given in terms of **odds**, a comparison of $P(E)$ with $P(E')$. For example, suppose $P(E) = 4/5$. Then $P(E') = 1 - 4/5 = 1/5$. These probabilities predict that E will occur 4 out of 5 times and E' will occur 1 out of 5 times. Then we say the *odds in favor* of E are 4 to 1.

Odds

The **odds in favor** of an event E are defined as the ratio of $P(E)$ to $P(E')$, or

$$\frac{P(E)}{P(E')}, \text{ where } P(E') \neq 0.$$

**EXAMPLE 10 Odds of Flat Feet**

About 1 in 7 children in the United States has persistent flat feet, which do not usually require treatment unless they cause pain. Find the odds that a child will have flat feet.

SOLUTION Let E be the event “flat feet.” Then E' is the event “no flat feet.” Since $P(E) = 1/7$, $P(E') = 6/7$. By the definition of odds, the odds in favor of flat feet are

$$\frac{1/7}{6/7} = \frac{1}{6}, \quad \text{written} \quad 1 \text{ to } 6, \text{ or } 1:6.$$

On the other hand, the odds that a child will not have flat feet, or the *odds against* flat feet, are

$$\frac{6/7}{1/7} = \frac{6}{1}, \quad \text{written} \quad 6 \text{ to } 1, \text{ or } 6:1. \quad \text{TRY YOUR TURN 8}$$

If the odds in favor of an event are, say, 3 to 5, then the probability of the event is $3/8$, while the probability of the complement of the event is $5/8$. (Odds of 3 to 5 indicate 3 outcomes in favor of the event out of a total of 8 possible outcomes.) This example suggests the following generalization.

If the odds favoring event E are m to n , then

$$P(E) = \frac{m}{m+n} \quad \text{and} \quad P(E') = \frac{n}{m+n}.$$

**EXAMPLE 11 Motorcycle Accidents**

Suppose the odds of dying while riding on a motorcycle are 1 to 1250. What is the probability of death while riding on a motorcycle?

SOLUTION The odds indicate chances of 1 out of 1251 ($1 + 1250 = 1251$) that one will die, so

$$P(\text{death}) = \frac{1}{1251}. \quad \text{TRY YOUR TURN 9}$$

Empirical Probability In many real-life problems, it is not possible to establish exact probabilities for events. Instead, useful approximations are often found by drawing on past experience. The next example shows one approach to such **empirical probabilities**.

**EXAMPLE 12 Injuries**

The following table lists the estimated number of injuries in the United States associated with recreation equipment. *Source: National Safety Council.*

Recreation Equipment Injuries	
Equipment	Number of Injuries
Bicycles	515,871
Skateboards	143,682
Trampolines	107,345
Playground climbing equipment	77,845
Swings or swing sets	59,144

YOUR TURN 8 If the probability that a male is left handed is $1/10$, find the odds that a male is left-handed.

YOUR TURN 9 If the odds of having twins are 3 to 97, find the probability of not having twins.

Find the probability that a randomly selected person whose injury is associated with recreation equipment was hurt on a trampoline.

SOLUTION We first find the total number of injuries. Verify that the amounts in the table sum to 903,887. The probability is then found by dividing the number of people injured on trampolines by the total number of people injured. Thus,

$$P(\text{Trampolines}) = \frac{107,345}{903,887} \approx 0.1188.$$

TRY YOUR TURN 10

YOUR TURN 10 In Example 12, find the probability the randomly selected person was hurt on a skateboard.

Probability Distribution A table listing each possible outcome of an experiment and its corresponding probability is called a **probability distribution**. The assignment of probabilities may be done in any reasonable way (on an empirical basis, as in the next example, or by theoretical reasoning) provided that it satisfies the following conditions.

Properties of Probability

Let S be a sample space consisting of n distinct outcomes, $s_1, s_2, \dots, s_n$. An acceptable probability assignment consists of assigning to each outcome s_i a number p_i (the probability of s_i) according to these rules.

1. The probability of each outcome is a number between 0 and 1, inclusive.

$$0 \leq p_1 \leq 1, \quad 0 \leq p_2 \leq 1, \dots, \quad 0 \leq p_n \leq 1$$

2. The sum of the probabilities of all possible outcomes is 1.

$$p_1 + p_2 + p_3 + \dots + p_n = 1$$

EXAMPLE 13 Causes of Death

There were 2,468,435 deaths in the United States in 2010 and they are listed, according to cause, in the following table. *Source: National Vital Statistics Report.*

Cause	Number of Deaths
Heart disease	597,689
Cancer	574,743
Cerebrovascular disease	129,476
Chronic lower respiratory diseases	138,080
Accidents	120,859
Pneumonia and influenza	50,097
Diabetes mellitus	69,071
All other causes	788,420

- (a) Construct a probability distribution for the probability that a death was caused by each category.

SOLUTION We can find the empirical probability that a death was caused by each category by dividing the number of deaths for each cause by the total number of deaths. Verify that the amounts in the table sum to 2,468,435. The probability that a randomly selected death was caused by an accident, for example, is $P(\text{death by accident}) = 120,859/2,468,435 \approx 0.049$. Similarly, we could divide each amount by

2,468,435, with the results (rounded to three decimal places) shown in the following table.

Cause	Probabilities
Heart disease	0.242
Cancer	0.233
Cerebrovascular disease	0.052
Chronic lower respiratory diseases	0.056
Accidents	0.049
Pneumonia and influenza	0.020
Diabetes mellitus	0.028
All other causes	0.319

Verify that this distribution satisfies the conditions for probability. Each probability is between 0 and 1, so the first condition holds. The probabilities in this table sum to 0.999. In theory, to satisfy the second condition, they should total 1.000, but this does not always occur when the individual numbers are rounded.

- (b) Find the probability that a death in the United States was caused by heart disease or cancer.

APPLY IT

SOLUTION The categories in the table are mutually exclusive simple events. Thus, to find the probability that a death is caused by heart disease or cancer, we use the union rule to calculate

$$P(\text{heart disease or cancer}) = 0.242 + 0.233 = 0.475$$

We could get this same result by summing the number of deaths by heart disease and cancer and dividing the total by 2,468,435.

Thus, almost half of all deaths were caused by heart disease or cancer.

A Venn diagram is often helpful in determining probabilities, as illustrated in the final example.



EXAMPLE 14 Health

A run-down college student has determined the probability that she will get a cold this semester is 0.47, the probability she will get the flu is 0.59, and the probability that she will get both is 0.31.

- (a) Find the probability that she will not get a cold *and* will not get the flu this semester.

SOLUTION Let C represent the event “gets a cold,” and F represent “gets the flu.” Place the given information on a Venn diagram, starting with 0.31 in the intersection of the regions C and F (see Figure 18). As stated earlier, event C has probability 0.47. Since 0.31 has already been placed inside the intersection of C and F ,

$$0.47 - 0.31 = 0.16$$

goes inside region C , but outside the intersection of C and F , that is, in the region $C \cap F'$. In the same way,

$$0.59 - 0.31 = 0.28$$

goes inside the region for F , and outside the overlap, that is, in the region $F \cap C'$.

Using regions C and F , the event we want is $C' \cap F'$. From the Venn diagram in Figure 18, the labeled regions have a total probability of

$$0.16 + 0.31 + 0.28 = 0.75.$$

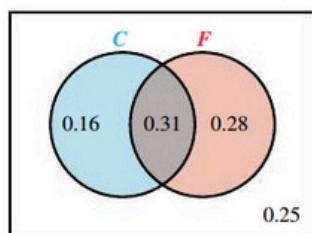


FIGURE 18

Since the entire region of the Venn diagram must have probability 1, the region outside C and F , or $C' \cap F'$, has probability

$$1 - 0.75 = 0.25.$$

The probability is 0.25 that she will not get a cold and will not get the flu this semester.

- (b) Find the probability that she will not get a cold or will not get the flu this semester.

SOLUTION The corresponding region, $C' \cup F'$, has probability

$$0.25 + 0.16 + 0.28 = 0.69,$$

or $1 - 0.31 = 0.69$, using the Complement Rule.

12.2 EXERCISES



1. What is the sample space for an experiment?
2. Define an event. What is a simple event?



Write sample spaces for the experiments in Exercises 3–8.

3. A month of the year is chosen for a wedding.
4. A student is asked how many questions she answered correctly on a recent 80-question test.
5. A patient must decide whether to have surgery, treat the problem with medicine, or wait six months to see if the problem resolves.
6. A record is kept each day for three days about whether a patient's blood pressure goes up or goes down.
7. A coin is tossed, and a die is rolled.
8. A box contains five balls, numbered 1, 2, 3, 4, and 5. A ball is drawn at random, the number on it recorded, and the ball replaced. The box is shaken, a second ball is drawn, and its number is recorded.

For the experiments in Exercises 9–14, write out the sample space S , choosing an S with equally likely outcomes, if possible. Then give the value of $n(S)$ and tell whether the outcomes in S are equally likely. Finally, write the indicated events in set notation.



9. A committee of 2 people is selected at random from 5 doctors: Alam, Bartolini, Chinn, Dickson, and Ellsberg.
 - a. Chinn is on the committee.
 - b. Dickson and Ellsberg are not both on the committee.
 - c. Both Alam and Chinn are on the committee.
10. Five states are being considered as the location for three new high-energy physics laboratories: California (CA), Colorado (CO), New Jersey (NJ), New York (NY), and Utah (UT). Three states will be chosen at random. Write elements of the sample space in the form (CA, CO, NJ).
 - a. All three states border an ocean.
 - b. Exactly two of the three states border an ocean.
 - c. Exactly one of the three states is west of the Mississippi River.



11. Slips of paper marked with the numbers 1, 2, 3, 4, and 5 are placed in a box. After being mixed, two slips are drawn simultaneously.
 - a. Both slips are marked with even numbers.
 - b. One slip is marked with an odd number and the other is marked with an even number.
 - c. Both slips are marked with the same number.
12. An unprepared student takes a three-question, true/false quiz in which he guesses the answers to all three questions, so each answer is equally likely to be correct or wrong.
 - a. The student gets three answers wrong.
 - b. The student gets exactly two answers correct.
 - c. The student gets only the first answer correct.
13. A coin is flipped until two heads appear, up to a maximum of four flips. (If three tails are flipped, the coin is still tossed a fourth time to complete the experiment.)
 - a. The coin is tossed four times.
 - b. Exactly two heads are tossed.
 - c. No heads are tossed.
14. One jar contains four balls, labeled 1, 2, 3, and 4. A second jar contains five balls, labeled 1, 2, 3, 4, and 5. An experiment consists of taking one ball from the first jar, and then taking a ball from the second jar.
 - a. The number on the first ball is even.
 - b. The number on the second ball is even.
 - c. The sum of the numbers on the two balls is 5.
 - d. The sum of the numbers on the two balls is 1.

A single fair die is rolled. Find the probabilities of each event.

15. Getting a 2
16. Getting an odd number
17. Getting a number less than 5
18. Getting a number greater than 2
19. Getting a 3 or a 4
20. Getting any number except 3

A card is drawn from a well-shuffled deck of 52 cards. Find the probability of drawing the following.

- 
- | | |
|------------------------|--------------------------|
| 21. A 9 | 22. A black card |
| 23. A black 9 | 24. The 9 of hearts |
| 25. A 2 or a queen | 26. A black 7 or a red 8 |
| 27. A red card or a 10 | 28. A spade or a king |

A jar contains 3 white, 4 orange, 5 yellow, and 8 black marbles. If a marble is drawn at random, find the probability that it is the following.

- 
- | | |
|---------------|----------------------|
| 29. White | 30. Yellow |
| 31. Not black | 32. Orange or yellow |
-  33. Define mutually exclusive events in your own words.
-  34. Explain the union rule for mutually exclusive events.

Decide whether the events in Exercises 35–38 are mutually exclusive.

- 
- 
- | |
|--|
| 35. Owning a dog and owning a smartphone. |
| 36. Being a biology major and being from Texas |
| 37. Being a teenager and being 70 years old |
| 38. Being one of the ten tallest people in the United States and being under 4 feet tall |

Two dice are rolled. Find the probabilities of rolling the given sums.

- 
- | | | | |
|------------------------|--------------------|--------------------------------|-------|
| 39. a. 2 | b. 4 | c. 5 | d. 6 |
| 40. a. 8 | b. 9 | c. 10 | d. 13 |
| 41. a. 9 or more | b. Less than 7 | c. Between 5 and 8 (exclusive) | |
| 42. a. Not more than 5 | b. Not less than 8 | c. Between 3 and 7 (exclusive) | |

Two dice are rolled. Find the probabilities of the following events.

- 
- | |
|---|
| 43. The first die is 3 or the sum is 8. |
| 44. The second die is 5 or the sum is 10. |

One card is drawn from an ordinary deck of 52 cards. Find the probabilities of drawing the following cards.

- 
- | |
|---|
| 45. a. A 9 or 10 |
| b. A red card or a 3 |
| c. A 9 or a black 10 |
| d. A heart or a black card |
| e. A face card or a diamond |
| 46. a. Less than a 4 (count aces as ones) |
| b. A diamond or a 7 |
| c. A black card or an ace |
| d. A heart or a jack |
| e. A red card or a face card |

Kristi Perez invites 13 relatives to a party: her mother, 2 aunts, 3 uncles, 2 brothers, 1 male cousin, and 4 female cousins. If the chances of any one guest arriving first are equally likely, find the probabilities that the first guest to arrive is as follows.

- 
- | |
|------------------------------|
| 47. a. A brother or an uncle |
| b. A brother or a cousin |
| c. A brother or her mother |
| 48. a. An uncle or a cousin |
| b. A male or a cousin |
| c. A female or a cousin |

The numbers 1, 2, 3, 4, and 5 are written on slips of paper, and 2 slips are drawn at random one at a time without replacement. Find the probabilities in Exercises 49 and 50.

- | |
|--|
| 49. a. The sum of the numbers is 9. |
| b. The sum of the numbers is 5 or less. |
| c. The first number is 2 or the sum is 6. |
| 50. a. Both numbers are even. |
| b. One of the numbers is even or greater than 3. |
| c. The sum is 5 or the second number is 2. |

 51. Define what is meant by odds.

 52. On page 134 of Roger Staubach's autobiography, *First Down, Lifetime to Go*, Staubach makes the following statement regarding his experience in Vietnam: Source: *First Down, Lifetime to Go*.

"Odds against a direct hit are very low but when your life is in danger, you don't worry too much about the odds."

Is this wording consistent with our definition of odds, for and against? How could it have been said so as to be technically correct?

A single fair die is rolled. Find the odds in favor of getting the results in Exercises 53–56.

- 
- | |
|--|
| 53. 3 |
| 54. 4, 5, or 6 |
| 55. 2, 3, 4, or 5 |
| 56. Some number less than 6 |
| 57. A marble is drawn from a box containing 3 yellow, 4 white, and 11 blue marbles. Find the odds in favor of drawing the following. |
| a. A yellow marble |
| b. A blue marble |
| c. A white marble |
| d. Not drawing a white marble |
| 58. Two dice are rolled. Find the odds of rolling the following. (Refer to Figure 17.) |
| a. A sum of 3 |
| b. A sum of 7 or 11 |
| c. A sum less than 5 |
| d. Not a sum of 6 |



Which of Exercises 59–64 are examples of empirical probability?

59. The probability of heads on 5 consecutive tosses of a coin
60. The probability that a person is allergic to penicillin
61. The probability of drawing an ace from a standard deck of 52 cards
62. The probability that a person will get lung cancer from smoking cigarettes
63. A surgeon's prediction that a patient has a 90% chance of a full recovery
64. A gambler's claim that on a roll of a fair die, $P(\text{even}) = 1/2$
65. What is a probability distribution?
66. What conditions must hold for a probability distribution to be acceptable?

An experiment is conducted for which the sample space is $S = \{s_1, s_2, s_3, s_4, s_5\}$. Which of the probability assignments in Exercises 67–72 are possible for this experiment? If an assignment is not possible, tell why.



67.	Outcomes	s_1	s_2	s_3	s_4	s_5
	Probabilities	0.09	0.32	0.21	0.25	0.13

68.	Outcomes	s_1	s_2	s_3	s_4	s_5
	Probabilities	0.92	0.03	0	0.02	0.03

69.	Outcomes	s_1	s_2	s_3	s_4	s_5
	Probabilities	$1/3$	$1/4$	$1/6$	$1/8$	$1/10$

70.	Outcomes	s_1	s_2	s_3	s_4	s_5
	Probabilities	$1/5$	$1/3$	$1/4$	$1/5$	$1/10$

71.	Outcomes	s_1	s_2	s_3	s_4	s_5
	Probabilities	0.64	-0.08	0.30	0.12	0.02

72.	Outcomes	s_1	s_2	s_3	s_4	s_5
	Probabilities	0.05	0.35	0.5	0.2	-0.3

One way to solve a probability problem is to repeat the experiment many times, keeping track of the results. Then the probability can be approximated using the basic definition of the probability of an event E : $P(E) = n(E)/n(S)$, where E occurs $n(E)$ times out of $n(S)$ trials of an experiment. This is called the Monte Carlo method of finding probabilities. If physically repeating the experiment is too tedious, it may be simulated using a random-number generator, available on most computers and scientific or graphing calculators. To simulate a coin toss or the roll of a die on the TI-84 Plus, change the setting to fixed decimal mode with 0 digits displayed, and enter `rand` or `rand*6+.5`, respectively. For a coin toss, interpret 0 as a head and 1 as a tail. In either case, the ENTER key can be pressed repeatedly to perform multiple simulations.



73. Suppose two dice are rolled. Use the Monte Carlo method with at least 50 repetitions to approximate the following probabilities. Compare with the results of Exercise 41.
 - a. $P(\text{the sum is 9 or more})$
 - b. $P(\text{the sum is less than 7})$



74. Suppose two dice are rolled. Use the Monte Carlo method with at least 50 repetitions to approximate the following probabilities. Compare with the results of Exercise 42.

- a. $P(\text{the sum is not more than 5})$
- b. $P(\text{the sum is not less than 8})$



75. Suppose three dice are rolled. Use the Monte Carlo method with at least 100 repetitions to approximate the following probabilities.

- a. $P(\text{the sum is 5 or less})$
- b. $P(\text{neither a 1 nor a 6 is rolled})$



76. Suppose a coin is tossed 5 times. Use the Monte Carlo method with at least 50 repetitions to approximate the following probabilities.

- a. $P(\text{exactly 4 heads})$
- b. $P(2 \text{ heads and 3 tails})$

Use Venn diagrams to work Exercises 77 and 78.



77. Suppose $P(E) = 0.26$, $P(F) = 0.41$, and $P(E \cap F) = 0.16$. Find the following.

- a. $P(E \cup F)$
- b. $P(E' \cap F)$
- c. $P(E \cap F')$
- d. $P(E' \cup F')$



78. Let $P(Z) = 0.42$, $P(Y) = 0.35$, and $P(Z \cup Y) = 0.59$. Find each probability.

- a. $P(Z' \cap Y')$
- b. $P(Z' \cup Y')$
- c. $P(Z' \cup Y)$
- d. $P(Z \cap Y')$



79. The student sitting next to you in class concludes that the probability of the ceiling falling down on both of you before class ends is $1/2$, because there are two possible outcomes—the ceiling will fall or not fall. What is wrong with this reasoning?

80. The following puzzler was given on the *Car Talk* radio program. Source: *Car Talk*, Feb 24, 2001.

"Three different numbers are chosen at random, and one is written on each of three slips of paper. The slips are then placed face down on the table. The objective is to choose the slip upon which is written the largest number. Here are the rules: You can turn over any slip of paper and look at the amount written on it. If for any reason you think this is the largest, you're done; you keep it. Otherwise you discard it and turn over a second slip. Again, if you think this is the one with the biggest number, you keep that one and the game is over. If you don't, you discard that one too. . . . The chance of getting the highest number is one in three. Or is it? Is there a strategy by which you can improve the odds?"

The answer to the puzzler is that you can indeed improve the probability of getting the highest number by the following strategy. Pick one of the slips of paper, and after looking at the number, throw it away. Then pick a second slip; if it has a larger number than the first slip, stop. If not, pick the third slip. Find the probability of winning with this strategy.*

*This is a special case of the famous Googol problem. For more details, see "Recognizing the Maximum of a Sequence" by John P. Gilbert and Frederick Mosteller, *Journal of the American Statistical Association*, Vol. 61, No. 313, March 1966, pp. 35–73.

-  **81.** The following description of the classic “Linda problem” appeared in the *New Yorker*: “In this experiment, subjects are told, ‘Linda is thirty-one years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice and also participated in antinuclear demonstrations.’ They are then asked to rank the probability of several possible descriptions of Linda today. Two of them are ‘bank teller’ and ‘bank teller and active in the feminist movement.’” Many people rank the second event as more likely. Explain why this violates basic concepts of probability. *Source: New Yorker.*
- 82.** Three unusual dice, A , B , and C , are constructed such that die A has the numbers 3, 3, 4, 4, 8, 8; die B has the numbers 1, 1, 5, 5, 9, 9; and die C has the numbers 2, 2, 6, 6, 7, 7.
- If dice A and B are rolled, find the probability that B beats A , that is, the number that appears on die B is greater than the number that appears on die A .
 - If dice B and C are rolled, find the probability that C beats B .
 - If dice A and C are rolled, find the probability that A beats C .
-  **d.** Which die is better? Explain.

LIFE SCIENCE APPLICATIONS

-  **83. Medical Survey** For a medical experiment, people are classified as to whether they smoke, have a family history of heart disease, or are overweight. Define events E , F , and G as follows.
- E : person smokes
 F : person has a family history of heart disease
 G : person is overweight
- Describe each event in words.
- G'
 - $F \cap G$
 - $E \cup G'$
-  **84. Medical Survey** Refer to Exercise 83. Describe each event in words.
- $E \cup F$
 - $E' \cap F$
 - $F' \cup G'$
- 85. Body Types** A study on body types gave the following results: 45% were short; 25% were short and overweight; and 24% were tall and not overweight. Find the probabilities that a person is the following.
- Overweight
 - Short, but not overweight
 - Tall and overweight
- 86. Color Blindness** Color blindness is an inherited characteristic that is more common in males than in females. If M represents male and C represents red-green color blindness, we use the relative frequencies of the incidences of males and red-green color blindness as probabilities to get

$$P(C) = 0.039, P(M \cap C) = 0.035, P(M \cup C) = 0.491.$$

Source: Parsons' Diseases of the Eye.

Find the following probabilities.

- $P(C')$
- $P(M)$
- $P(M')$
- $P(M' \cap C')$
- $P(C \cap M')$
- $P(C \cup M')$

- 87. Genetics** Gregor Mendel, an Austrian monk, was the first to use probability in the study of genetics. In an effort to understand the mechanism of character transmittal from one generation to the next in plants, he counted the number of occurrences of various characteristics. Mendel found that the flower color in certain pea plants obeyed this scheme:

Pure red crossed with pure white produces red.

From its parents, the red offspring received genes for both red (R) and white (W), but in this case red is *dominant* and white *recessive*, so the offspring exhibits the color red. However, the offspring still carries both genes, and when two such offspring are crossed, several things can happen in the third generation. The table below, which is called a *Punnett square*, shows the equally likely outcomes.

		Second Parent	
		R	W
First Parent	R	RR	RW
	W	WR	WW

Use the fact that red is dominant over white to find the following. Assume that there are an equal number of red and white genes in the population.

- $P(\text{a flower is red})$
 - $P(\text{a flower is white})$
- 88. Genetics** Mendel found no dominance in snapdragons, with one red gene and one white gene producing pink-flowered offspring. These second-generation pinks, however, still carry one red and one white gene, and when they are crossed, the next generation still yields the Punnett square from Exercise 87. Find each probability.
- $P(\text{red})$
 - $P(\text{pink})$
 - $P(\text{white})$
- (Mendel verified these probability ratios experimentally and did the same for many characteristics other than flower color. His work, published in 1866, was not recognized until 1890.)
- 89. Genetics** In most animals and plants, it is very unusual for the number of main parts of the organism (such as arms, legs, toes, or flower petals) to vary from generation to generation. Some species, however, have *meristic variability*, in which the number of certain body parts varies from generation to generation. One researcher studied the front feet of certain guinea pigs and produced the following probabilities. *Source: Genetics.*

$$P(\text{only four toes, all perfect}) = 0.77$$

$$P(\text{one imperfect toe and four good ones}) = 0.13$$

$$P(\text{exactly five good toes}) = 0.10$$

Find the probability of each event.

- a. No more than four good toes
- b. Five toes, whether perfect or not



90. **Doctor Visit** The probability that a visit to a primary care physician's (PCP) office results in neither lab work nor referral to a specialist is 35%. Of those coming to a PCP's office, 30% are referred to specialists and 40% require lab work. Determine the probability that a visit to a PCP's office results in both lab work and referral to a specialist. Choose one of the following. (Hint: Use the union rule for probability.) *Source: Society of Actuaries.*

- a. 0.05 b. 0.12 c. 0.18
- d. 0.25 e. 0.35



91. **Shoulder Injuries** Among a large group of patients recovering from shoulder injuries, it is found that 22% visit both a physical therapist and a chiropractor, whereas 12% visit neither of these. The probability that a patient visits a chiropractor exceeds by 0.14 the probability that a patient visits a physical therapist. Determine the probability that a randomly chosen member of this group visits a physical therapist. Choose one of the following. (Hint: Use the union rule for probability, and let $x = P(\text{patient visits a physical therapist})$.) *Source: Society of Actuaries.*

- a. 0.26 b. 0.38 c. 0.40
- d. 0.48 e. 0.62



92. **Health Plan** An insurer offers a health plan to the employees of a large company. As part of this plan, the individual employees may choose exactly two of the supplementary coverages A, B, and C, or they may choose no supplementary coverage. The proportions of the company's employees that choose coverages A, B, and C are $\frac{1}{4}$, $\frac{1}{3}$, and $\frac{5}{12}$, respectively. Determine the probability that a randomly chosen employee will choose no supplementary coverage. Choose one of the following. (Hint: Draw a Venn diagram with three sets, and let $x = P(A \cap B)$. Use the fact that 4 of the 8 regions in the Venn diagram have a probability of 0.) *Source: Society of Actuaries.*

- a. 0 b. $\frac{47}{144}$ c. $\frac{1}{2}$
- d. $\frac{97}{144}$ e. $\frac{7}{9}$

93. **U.S. Population** The projected U.S. population (in thousands) by race in 2020 and 2050 is given in the table. *Source: Bureau of the Census.*

Race	2020	2050
White	207,393	207,901
Hispanic	52,652	96,508
Black	41,538	53,555
Asian and Pacific Islander	18,557	32,432
Other	2602	3535

Find the probability that a randomly selected person in the given year is of the race specified.

- a. Hispanic in 2020 b. Hispanic in 2050
- c. Black in 2020 d. Black in 2050

OTHER APPLICATIONS

94. **Employment** The table shows the projected probabilities of a worker employed by different occupational groups in 2018. *Source: U.S. Department of Labor.*

Occupation	Probability
Management and business	0.1047
Professional	0.2182
Service	0.2024
Sales	0.1015
Office and administrative support	0.1560
Farming, fishing, forestry	0.0061
Construction	0.0531
Production	0.0585
Other	0.0995

If a worker in 2018 is selected at random, find the following.

- a. The probability that the worker is in sales or service.
- b. The probability that the worker is not in construction.
- c. The odds in favor of the worker being in production.

95. **Labor Force** The 2008 and the 2018 (projected) civilian labor forces by age are given in the following table. *Source: U.S. Department of Labor.*

Age (in years)	2008 (in millions)	2018 (in millions)
16 to 24	22.0	21.1
25 to 54	104.4	105.9
55 and over	27.9	39.8
Total	154.3	166.8

- a. In 2008, find the probability that a member of the civilian labor force is age 55 or older.
- b. In 2018, find the probability that a member of the civilian labor force is age 55 or over.
- c. What do these projections imply about the future civilian labor force?

96. **Labor Force** The following table gives the 2018 projected civilian labor force probability distribution by age and gender. *Source: U.S. Department of Labor.*

Age	Male	Female	Total
16–24	0.066	0.061	0.127
25–54	0.343	0.291	0.634
55 and over	0.122	0.117	0.239
Total	0.531	0.469	1.000

Find the probability that a randomly selected worker is the following.

- Female and 16 to 24 years old
- 16 to 54 years old
- Male or 25 to 54 years old
- Female or 16 to 24 years old



97. **Book of Odds** The following table gives the probabilities that a particular event will occur. Convert each probability to the odds in favor of the event. *Source: The Book of Odds.*

Event	Probability for the Event
An NFL pass will be intercepted.	0.03
A U.S. president owned a dog during his term in office.	0.65
A woman owns a pair of high heels.	0.61
An adult smokes.	0.21
A flight is cancelled.	0.02

98. **Book of Odds** The following table gives the odds that a particular event will occur. Convert each odd to the probability that the event will occur. *Source: The Book of Odds.*

Event	Odds for the Event
A Powerball entry will win the jackpot.	1 to 195,199,999
An adult will be struck by lightning during a year.	1 to 835,499
An adult will file for personal bankruptcy during a year.	1 to 157.6
A person collects stamps.	1 to 59.32



99. **Civil War** Estimates of the Union Army's strength and losses for the battle of Gettysburg are given in the following table, where *strength* is the number of soldiers immediately preceding the battle and *loss* indicates a soldier who was killed, wounded, captured, or missing. *Source: Regimental Strengths and Losses of Gettysburg.*

Unit	Strength	Loss
I Corps (Reynolds)	12,222	6059
II Corps (Hancock)	11,347	4369
III Corps (Sickles)	10,675	4211
V Corps (Sykes)	10,907	2187
VI Corps (Sedgwick)	13,596	242
XI Corps (Howard)	9188	3801
XII Corps (Slocum)	9788	1082
Cavalry (Pleasanton)	11,851	610
Artillery (Tyler)	2376	242
Total	91,950	22,803

- Find the probability that a randomly selected union soldier was from the XI Corps.
- Find the probability that a soldier was lost in the battle.
- Find the probability that a I Corps soldier was lost in the battle.
- Which group had the highest probability of not being lost in the battle?
- Which group had the highest probability of loss?
- Explain why these probabilities vary.



100. **Civil War** Estimates of the Confederate Army's strength and losses for the battle of Gettysburg are given in the following table, where *strength* is the number of soldiers immediately preceding the battle and *loss* indicates a soldier who was killed, wounded, captured, or missing. *Source: Regimental Strengths and Losses at Gettysburg.*

Unit	Strength	Loss
I Corps (Longstreet)	20,706	7661
II Corps (Ewell)	20,666	6603
III Corps (Hill)	22,083	8007
Cavalry (Stuart)	6621	286
Total	70,076	22,557

- Find the probability that a randomly selected confederate soldier was from the III Corps.
- Find the probability that a confederate soldier was lost in the battle.
- Find the probability that a I Corps soldier was lost in the battle.
- Which group had the highest probability of not being lost in the battle?
- Which group had the highest probability of loss?

101. **Perceptions of Threat** Research has been carried out to measure the amount of intolerance that citizens of Russia have for left-wing Communists and right-wing Fascists, as indicated in the table below. Note that the numbers are given as percents and each row sums to 100 (except for rounding). *Source: Political Research Quarterly.*

Russia	None at All	Don't Know	Not Very		
			Much	Somewhat	Extremely
Left-Wing Communists	47.8	6.7	31.0	10.5	4.1
Right-Wing Fascists	3.0	3.2	7.1	27.1	59.5